Delegate manual

Systematic reviews & meta-analysis — a take-home reference

To keep, and to use on your own reviews

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How to use this manual

The workbook you used on the day held the tasks. This manual is the reference you keep. It sets out the workflow stage by stage, points you to the definitive standards and tools, maps every pod to the relevant chapter of the Cochrane Handbook, and gathers a glossary and further reading. When you run your own review, start here.

The single most important resource is the Cochrane Handbook for Systematic Reviews of Interventions (currently version 6.5, 2024). It is free online and is the reference the whole field works to. Wherever this manual shows a purple "Cochrane Handbook" pointer, that is where to go deeper.

What is a systematic review?

A systematic review uses pre-specified, reproducible methods to find, appraise and combine all the studies that address a focused question. It differs from a narrative review in that every step — the question, the search, the selection, the appraisal, the synthesis — is planned in advance and documented so that someone else could repeat it and reach the same answer. A meta-analysis is the optional statistical step of pooling results quantitatively.

The value is twofold: it reduces bias by making the method explicit, and it increases precision by combining evidence. The cost is discipline — and that discipline is exactly what the standards and tools in this manual exist to support.

The workflow, stage by stage

1. Frame the question

Turn a real-world uncertainty into an answerable PICO question, write a protocol, and register it on PROSPERO before you start. Pre-specification is your main defence against bias.

Key standards / tools: PICO, PROSPERO **Cochrane Handbook:** Ch 2, Ch 3

2. Search

Build a sensitive, reproducible search across several databases, using controlled vocabulary (MeSH) and free-text terms. Record every line so it can be re-run; have a librarian check it.

Key standards / tools: Ovid/MEDLINE, Embase, CENTRAL **Cochrane Handbook:** Ch 4

3. Screen

Remove duplicates, then screen titles and abstracts against the criteria — ideally two reviewers, blinded, reconciling conflicts. Rayyan makes this practical and free.

Key standards / tools: Rayyan, dual screening **Cochrane Handbook:** Ch 4

4. Assess risk of bias

Judge each included study with the right tool for its design: RoB 2 for randomised trials, ROBINS-I for non-randomised studies. Risk of bias is a structured judgement, not a score.

Key standards / tools: RoB 2, ROBINS-I **Cochrane Handbook:** Ch 7, Ch 8, Ch 25

5. Extract data

Design and version-control an extraction form; two reviewers extract independently and reconcile. Record the source of every number. This is the stage that most needs human care.

Key standards / tools: Extraction template **Cochrane Handbook:** Ch 5, Ch 6

6. Synthesise

Choose an effect measure, prepare the data, and pool with a random-effects model where appropriate. Read the forest plot: the pooled estimate and its uncertainty.

Key standards / tools: R + metafor, forest plot **Cochrane Handbook:** Ch 6, Ch 9, Ch 10

7. Explore heterogeneity

Quantify inconsistency (Q, I^2) and, where planned, use subgroup analysis or meta-regression to explain it. Sometimes the heterogeneity is the finding.

Key standards / tools: Meta-regression **Cochrane Handbook:** Ch 10

8. Check for reporting bias & test robustness

Look for small-study effects with a funnel plot, and run pre-specified sensitivity analyses (e.g. leave-one-out). Weigh evidence by size and quality.

Key standards / tools: Funnel plot, sensitivity analysis **Cochrane Handbook:** Ch 13, Ch 10

9. Rate certainty & interpret

Use GRADE to rate the certainty of the evidence, summarise it, and interpret honestly — what the review can and cannot conclude.

Key standards / tools: GRADE, Summary of findings **Cochrane Handbook:** Ch 14, Ch 15

10. Beyond pairwise synthesis

When the question needs more: network meta-analysis to compare many treatments, IPD meta-analysis to use raw data, and qualitative synthesis for experience and meaning.

Key standards / tools: NMA, IPD, qualitative synthesis **Cochrane Handbook:** Ch 11, Ch 26, Ch 21, Ch 12

Key standards & tools

The essential references, and where to find each one.

Standard / tool	What it is	Where to find it
Cochrane Handbook	The definitive methods reference for systematic reviews of interventions (v6.5, 2024).	cochrane.org/handbook
PRISMA 2020	Reporting standard — the checklist and flow diagram every review should follow.	prisma-statement.org
PROSPERO	International prospective register of systematic reviews — register your protocol before you start.	crd.york.ac.uk/prospero
MECIR	Methodological Expectations of Cochrane Intervention Reviews — the conduct standards.	community.cochrane.org/mecir-manual
RoB 2	Risk-of-bias tool for randomised trials (five domains).	riskofbias.info
ROBINS-I	Risk-of-bias tool for non-randomised studies of interventions.	riskofbias.info
GRADE / GRADEpro	Framework and software for rating the certainty of a body of evidence.	gradeworkinggroup.org · gradepro.org
Rayyan	Free web tool for collaborative title/abstract screening.	rayyan.ai
R + metafor	Free statistical environment and the metafor package for meta-analysis.	r-project.org · metafor-project.org
Cochrane Library	Where Cochrane reviews are published; check for free access in your country.	cochranelibrary.com

The Cochrane Handbook, mapped to the day

Every pod maps onto the Handbook. Use this to go deeper on anything from the workshop. Chapters are from version 6.5 (2024).

Pod	Cochrane Handbook chapter(s)
Pod 1 · What & why	Ch 1 Starting a review; Ch II Planning a review
Pod 2 · Question, protocol, PROSPERO	Ch 2 Scope & questions; Ch 3 Inclusion criteria
Pod 3 · Searching	Ch 4 Searching for and selecting studies
Pod 4 · Screening (Rayyan)	Ch 4 Selecting studies
Pod 5 · Risk of bias	Ch 7 Considering bias; Ch 8 RoB 2 (RCTs); Ch 25 ROBINS-I (non-randomised)
Pod 6 · Responsible AI	No dedicated chapter — an emerging area; follow Cochrane's evolving guidance
Pod 7 · Data extraction	Ch 5 Collecting data; Ch 6 Effect measures
Pod 8 · Build a meta-analysis	Ch 6 Effect measures; Ch 9 Preparing for synthesis; Ch 10 Meta-analyses
Pod 9 · Heterogeneity & meta-regression	Ch 10 Analysing data and undertaking meta-analyses
Pod 10 · Publication bias & sensitivity	Ch 13 Bias due to missing results; Ch 10 (sensitivity analyses)
Pod 11 · IPD, NMA & Bayesian	Ch 11 Network meta-analyses; Ch 26 Individual participant data
Pod 12 · Qualitative & pipeline	Ch 21 Qualitative evidence; Ch 12 Synthesis using other methods
Cross-cutting · certainty & interpretation	Ch 14 Summary of findings & GRADE; Ch 15 Interpreting results

The datasets you used

- Zinc for childhood diarrhoea (simulated, patient-level) — build the meta-analysis from raw data. Pooled RR ≈ 0.71.
- BCG vaccine and TB (real; Colditz et al., 1994) — pooling, heterogeneity and a meta-regression on latitude. Pooled RR ≈ 0.49, $I^2 \approx 92\%$.
- Magnesium in acute MI (simulated cautionary tale) — small trials suggest benefit; the large trial overturns it.

All three, with their R scripts and preview plots, are in your Datasets and Exercises folders.

Responsible AI — the principles

- A named human is accountable for the review. AI assists; it does not sign off.
- Assume outputs can be confidently wrong (hallucinations). Verify against sources.
- Use AI where it is strong (drafting, translating, validating); guard the weak stages (raw numerical extraction).
- Report exactly how AI was used, so the review can be appraised and reproduced.
- Keep the originals. Efficiency never overrides accuracy.

Glossary

Term	Meaning
Risk ratio (RR)	Ratio of the risk of an event in one group to the risk in another. RR = 1 means no difference.
Odds ratio (OR)	Ratio of the odds of an event between groups. Approximates RR for rare outcomes; exaggerates it for common ones.
Confidence interval (CI)	The range within which the true effect plausibly lies; a 95% CI crossing 1 (for RR/OR) means the result is not statistically significant.
Fixed vs random effects	Fixed-effect assumes one true effect shared by all studies; random-effects allows the true effect to vary between studies.
Heterogeneity	Variation in results across studies beyond chance — clinical, methodological or statistical.
I^2 and Q	Q (Chi^2) tests for heterogeneity; I^2 is the percentage of variation beyond chance. Read them together, and cautiously with few studies.
tau² (τ^2)	The estimated variance of the true effects between studies in a random-effects model.
Meta-regression	Regressing the effect estimate on a study-level characteristic (e.g. latitude) to explain heterogeneity.
Forest plot	The classic display: each study's effect and CI, with a diamond for the pooled estimate.
Funnel plot	Effect size against precision; asymmetry can signal small-study effects or publication bias.
Publication bias	Distortion when studies are published (or not) depending on their results.
Sensitivity analysis	Re-running the analysis under different assumptions (e.g. leave-one-out) to test how robust the result is.
PICO	Population, Intervention, Comparator, Outcome — the frame for an answerable review question.
IPD meta-analysis	Meta-analysis using the raw individual-participant data rather than published summaries.
Network meta-analysis (NMA)	Comparing three or more treatments at once by combining direct and indirect evidence.
GRADE	A framework for rating the certainty of a body of evidence (high to very low).
Continuity correction	A small value added to zero cells so a ratio can be computed; the choice can affect the result.

Further reading & free resources

Go deeper

– Cochrane Handbook for Systematic Reviews of Interventions, v6.5 (2024): cochrane.org/handbook

– PRISMA 2020 statement and checklist: prisma-statement.org

– PROSPERO registration: crd.york.ac.uk/prospero

– GRADE handbook and working group: gradeworkinggroup.org

– metafor package (meta-analysis in R), with worked examples: metafor-project.org

Free & low-bandwidth resources

- Cochrane Interactive Learning and the Handbook are free online; check whether your country has free one-click access to the Cochrane Library.
- Rayyan, R and metafor are all free — no licence cost, which matters where subscriptions are limited.
- RoB 2 and ROBINS-I tools, templates and guidance are free at riskofbias.info.
- Prefer downloading materials once and working offline where connectivity is variable; keep original PDFs alongside any Markdown you convert.